

ALGEBRAIC FRACTIONS, (MULTIPLYING AND DIVIDING ALGEBRAIC

Name: _____

Date: _____

LO: I can multiply and divide algebraic fractions and simplify the result.

Multiplying and dividing algebraic fractions

To **multiply** algebraic fractions, multiply numerators together and denominators together. To **divide**, flip the second fraction and multiply.

Strategy: Factorise first where possible, cancel common factors, then multiply what remains.

Quick examples

●	$\frac{2}{x} \times \frac{x}{3} = \frac{2}{3}$	cancel x
●	$\frac{x}{3} \div \frac{x}{6} = \frac{x}{3} \times \frac{6}{x} = 2$	flip and multiply
●	$\frac{x+1}{x} \times \frac{x}{x+2} = \frac{x+1}{x+2}$	cancel x
●	$\frac{2}{x} \times \frac{3}{x} = \frac{6}{x^2}$	must multiply denominators too
●	$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$	must flip second fraction

Key idea

Multiply: top $\times$ top, bottom $\times$ bottom. Divide: flip the second fraction then multiply. Always cancel common factors.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

KFC: Keep, Flip, Change

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - simple multiplication

Simplify $\frac{3}{x} \times \frac{x}{5}$

Multiply numerators: $3 \times x = 3x$
 Multiply denominators: $x \times 5 = 5x$
 Cancel x: $\frac{3x}{5x} = \frac{3}{5}$

$$\frac{3}{5}$$

Cancel any common factors between numerator and denominator.

Example 2 - division

Simplify $\frac{4}{x} \div \frac{2}{x}$

Flip second fraction: $\frac{4}{x} \times \frac{x}{2}$
 Multiply: $\frac{4x}{2x}$
 Cancel: $\frac{4x}{2x} = 2$

$$= 2$$

Keep the first, flip the second, then multiply.

Example 3 - factorising first

Simplify $\frac{x^2+x}{x+1} \times \frac{2}{x}$

Factorise: $x^2+x = x(x+1)$
 Rewrite: $\frac{x(x+1)}{x+1} \times \frac{2}{x}$
 Cancel (x+1) and x: $= 2$

$$= 2$$

Always factorise before cancelling to spot common factors.

Example 4 - harder division

Simplify $\frac{x+3}{x-1} \div \frac{x+3}{x+2}$

Flip: $\frac{x+3}{x-1} \times \frac{x+2}{x+3}$
 Cancel (x+3)
 $\frac{x+2}{x-1}$

$$\frac{x+2}{x-1}$$

After flipping, cancel the common factor (x+3).

YOUR TURN – PRACTICE (deliberate practice)

1. Simple multiplication

Simplify each product.

- a) $\frac{2}{x} \times \frac{x}{4}$
- b) $\frac{3}{a} \times \frac{a}{9}$
- c) $\frac{x}{5} \times \frac{10}{x}$
- d) $\frac{4x}{3} \times \frac{6}{2x}$

2. Simple division

Simplify each quotient.

- a) $\frac{6}{x} \div \frac{3}{x}$
- b) $\frac{x}{4} \div \frac{x}{8}$
- c) $\frac{5x}{2} \div \frac{10x}{3}$
- d) $\frac{3}{2x} \div \frac{9}{4x}$

3. With factorising

Factorise where necessary then simplify.

- a) $\frac{x^2}{x+1} \times \frac{x+1}{x}$
- b) $\frac{2x+4}{x} \times \frac{x}{x+2}$
- c) $\frac{x^2-x}{3} \times \frac{6}{x-1}$
- d) $\frac{x^2+3x}{x+3} \div \frac{x}{2}$

4. Harder expressions

Simplify fully.

- a) $\frac{x+2}{x-1} \times \frac{x-1}{x+5}$
- b) $\frac{x+4}{3x} \div \frac{x+4}{6x}$
- c) $\frac{x^2-4}{x} \times \frac{x}{x+2}$
- d) $\frac{x^2-9}{x+1} \div \frac{x-3}{x+1}$

5. Mixed operations

Simplify each expression fully.

- a) $\frac{2x}{5} \times \frac{15}{4x}$
- b) $\frac{x+1}{x-2} \div \frac{x+1}{x+3}$
- c) $\frac{x^2-1}{x} \times \frac{x}{x+1}$
- d) $\frac{3x+6}{4} \div \frac{x+2}{8}$
- e) $\frac{x^2+5x+6}{x+1} \times \frac{x+1}{x+3}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To divide fractions, flip the second and multiply ☐

Correct answer: _____

Reason: _____

b) $\frac{2}{x} \times \frac{3}{x} = \frac{6}{x}$ ☐

Correct answer: _____

Reason: _____

c) $\frac{x+1}{x+2} \times \frac{x+2}{x+3} = \frac{x+1}{x+3}$ ☐

Correct answer: _____

Reason: _____

d) $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{c}{d}$ ☐

Correct answer: _____

Reason: _____

e) You should factorise before cancelling ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Simplify $\frac{x^2-4}{x^2+5x+6} \times \frac{x+3}{x-2}$. Factorise all quadratics fully before cancelling.

Expression: _____

Simplified: _____

b) Show that $\frac{x^2-9}{x+1} \div \frac{x^2-6x+9}{x^2+x}$ simplifies to $\frac{x(x+3)}{x-3}$.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

